

DIS1 through DIS3. Resistors R10 through R16 are used for limiting the current.

CD4553 provides BCD counts on D0 through D3 (which are converted into segment data by CD4511) and display-enable signal from DIS1 through DIS3 pins simultaneously in time-division-multiplexed mode for displaying a particular number on the 7-segment display unit.

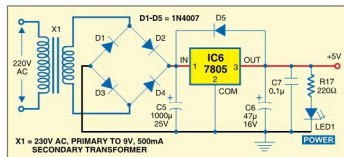


Fig. 3: Power supply circuit

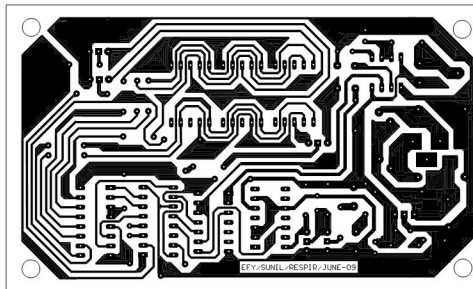


Fig. 4: Actual-size, single-side PCB for the digital respiration rate meter

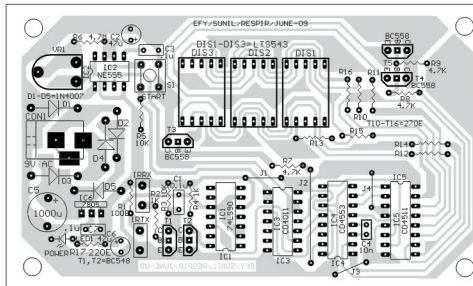


Fig. 5: Component layout for the PCB

play unit. Segment data and display-enable pulse for display are refreshed more than 25 times every second. Thus, the display appears to be continuous, even though DIS1 through DIS3 light up one by one.

When pin 2 of CD4553 goes low to drive transistor T5 into saturation and provides ground path to common-cathode pin 3 or pin 8 of DIS1, pins 1 and 15 of CD4553 also go low and transistors T4 and T3 drive common-cathode pins 3 or 8 of displays DIS2 and DIS3.

Fig. 3 shows the circuit of the power supply. The 230V AC mains is stepped down by transformer X1 to deliver the secondary output of 9V, 500 mA. The transformer output is rectified by a full-wave bridge rectifier comprising diodes D1 through D4, filtered by capacitor C5 and then regulated by IC 7805 (IC6). Capacitors C6 and C7 bypass the ripples present in the regulated supply. LED1 acts as a power-on indicator and resistor R17 limits its current through LED1. As a capacitor above 10 µF is connected to the output of the regulator IC, diode D5 is used to provide protection to the regulator IC in case its input shorts to ground.

Construction

An actual-size, single-side PCB for the respiration rate meter is shown in Fig. 4 and its component layout in Fig. 5. Fig. 6 shows the physical assembly for the respiration rate meter. The interface assembly simply makes use of a nose mask, which is readily available in medical shops and mainly used in hospitals for supply of oxygen to the patients for respiration.

Take a 36cm long transparent glass tube with inner diameter of 5 mm. Insert a